

Physical activity containing behavioural interventions for adults living with modifiable chronic non-communicable diseases in Africa: a systematic mixed-studies review

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Background: Physical activity improves physiological, cognitive and psychosocial functioning in chronic non-communicable diseases (NCDs). This study reviewed papers on the effects and patients' experiences of physical activity interventions for chronic NCDs in Africa.

Methods: We conducted a systematic review of clinical and qualitative studies by searching eight bibliographic databases and grey literature until 19 April 2017. The mixed-methods appraisal and Cochrane Collaboration's tools were used for quality and risk of bias assessments. Three-stage sequential explanatory syntheses were done.

Results: One randomized controlled trial (RCT), two non-controlled before and after studies and two qualitative studies of diabetic South African and Reunion patients were included. Exercise and sports unrelated to home and occupational activities were increased in the long term (1 year, moderate quality evidence) and short term immediately after a 4-week intervention (low quality evidence). There was conflicting evidence of intervention effects on home and occupational physical activities. Behaviour-change techniques improving chronic disease knowledge, addressing environmental barriers and stimulating/supporting physical activity were important to patients. Procedure-related components—health professional training and adequate health facilities—were important to patients, but were not addressed.

Conclusion: High quality RCTs are needed to confirm the intervention components for improving physical activity for chronic NCD management in Africa.

Keywords: Africa, behaviour change, chronic non-communicable diseases, diabetes, physical activity

Introduction

Eighty per cent of chronic non-communicable disease (NCD) mortality occurs in developing countries, particularly in Africa.¹ The most common NCDs include cardiovascular diseases, diabetes, cancer and chronic respiratory diseases,² with positive correlation between NCDs and bone and joint disorders.³ Obesity is a common risk factor for NCDs, and a multifactorial link exists between obesity and chronic pain, including genetic, metabolic, psychological and mechanical factors.^{4–6} Additionally, chronic pain and exercise incapacity are associated with NCDs;⁷ for example, people

with chronic back pain have a greater risk of all-cause and cardiovascular mortality.⁸

Evidence-based treatment guidelines for NCDs recommend physical activity to improve physiological, cognitive, emotional, social and psychosocial functioning, alter health beliefs, increase acceptance of chronic disease and reduce disability.^{9–19} Systematic reviews suggest that specific interventions can increase physical activity.²⁰ This reverses or reduces chronic disease progression and premature deaths,²¹ and can reduce the pain and disability associated with chronic musculoskeletal pain.^{18,22,23}

For adults aged 18–64 years, the WHO recommends 150 min of moderate-intensity aerobic physical activity or at least 75 min of vigorous-intensity aerobic physical activity throughout the week, or an equivalent combination of moderate-intensity and vigorous-intensity physical activity; plus muscle strengthening activities involving major muscle groups done on two or more days a week.²⁴ However, independent of gender, ethnicity and income, more than half of all adults in Africa are inactive.²⁵

NCDs require ongoing management over years or decades, and need a coordinated health care model centred on patients' circumstances to facilitate self-management skills and behaviour change.²⁶ However, behaviour patterns are deeply embedded in people's cultural, social and economic characteristics,²⁷ and hence partly controlled by intrapsychic and external factors,²⁸ irrespective of the chronic condition.²⁹

For the purposes of this study, behaviour-change interventions are coordinated activities designed to change particular behaviours.²⁸ They are often complex, comprising many interacting components.^{30,31} Behaviour-change intervention components are the techniques that constitute the active ingredients of the intervention, and the procedures used to deliver those techniques.³¹ Behaviour change techniques (BCTs) are replicable components of an intervention designed to alter/redirect causal processes that regulate behaviour with minimum delivery specifications that allow their identification.^{31,32} They are useful for identifying and standardizing the reporting of complex interventions in systematic reviews.^{31,32} Procedures include who delivers the intervention, to whom, how often, for how long, in what format and in what context.^{31,33} Physical activity is any bodily movement resulting from the contraction of skeletal muscles that leads to energy expenditure.³⁴

No systematic review has examined the effects of behaviour change interventions on physical activity behaviour of people living with chronic NCDs in Africa; patients' experiences of these interventions; and to what extent interventions addressed patients' concerns. These are the objectives of this study.

Methods

Search strategy and selection criteria

This systematic review was registered with the International Prospective Register of Systematic Reviews (CRD42015016084) and the protocol published.³⁵ Following PRISMA guidelines,³⁶ we searched MEDLINE, EMBASE, PsycINFO, CENTRAL, CINAHL, Web of Science, African Index Medicus (AIM) and AFROLIB (WHO's regional office database for Africa) from their inception dates up to 19 April 2017.

We used a combination of Medical Subject Headings, free text terms and word variants for hypertension, diabetes, obesity and chronic musculoskeletal pain conditions, African search filter,³⁷ word variants for physical activity-related therapies and the search filters for three study types—clinical trials, qualitative and observational studies (supplemental file).³⁵ We also searched grey literature as described in the protocol. These conditions were selected because they are the most prevalent in Africa,^{1,38} and may be the most easily modified by physical activity. Lung diseases were not included because it was

envisaged that a significant proportion may be secondary to communicable diseases, such as tuberculosis.

The reference lists of relevant systematic reviews were appraised for potentially eligible studies.

There was no restriction by language or intervention settings. Studies were selected if they involved adults (≥ 18 years) living in Africa with hypertension, diabetes, obesity or chronic musculoskeletal pain.

There were no restrictions on study design, comparators and timing. Clinical studies (randomized controlled trials [RCTs], non-controlled before and after [B/A] studies and mixed-methods B/A studies) were included if they investigated the effects of physical activity interventions on physical activity behaviour. Qualitative studies were included if they explored experiences and perspectives of individuals of these interventions. Physical activity behaviour change was the primary outcome evaluated in this review. We excluded studies of African patients not living in Africa; studies without a physical activity intervention, physical activity behavioural assessment or any primary data; and duplicate publications.

We selected studies in two stages. In the first stage, CNI-C (first author) screened titles and abstracts using the eligibility criteria to identify potentially relevant articles. This was cross-checked by ELG (last author), APK (second author) and IOS (third author). In the second stage, screening of full articles was done independently by CNI-C, ELG and IOS. Disagreements were resolved by consultation with APK. Further details are in the review protocol³⁵ and the flow chart (Figure 1).

Data extraction and quality assessment

This was done by CNI-C, and cross-checked independently by ELG and IOS. Disagreements were resolved by consultation with APK. The data collection process is detailed in the protocol.³⁵ Intervention descriptions were recorded as described by the primary authors and subsequently coded with the taxonomy of hierarchically clustered BCTs³⁰ using the definition of Michie, Abraham³¹ and Whitlock.³³ Consensus agreement in coding was achieved via group discussion of the multidisciplinary review team.

We assessed the quality of all studies with the Mixed Methods Appraisal Tool (MMAT)³⁹ as described in the review protocol.³⁵ For RCTs, quality criteria included appropriate random sequence generation/clear description of randomization, allocation concealment or blinding, complete outcome data ($\geq 80\%$) and low drop-out rate ($< 20\%$). For non-controlled B/A studies, the criteria were low selection bias in recruiting participants representative of the population; valid and reliable outcome measures; accounting for confounding and co-variate variables; complete outcome data ($\geq 80\%$); acceptable response rate ($\geq 60\%$) and follow-up rate. For mixed-methods B/A studies, the overall quality score is the lowest score of the study components. For qualitative studies, the criteria were adequate/relevant sources of qualitative data, qualitative data analysis addressed research objectives, consideration was given to how results were related to context, and consideration was given to how findings related to researcher's influence.³⁹

A total score for each study was based on the number of criteria met (Tables 1 and 3).

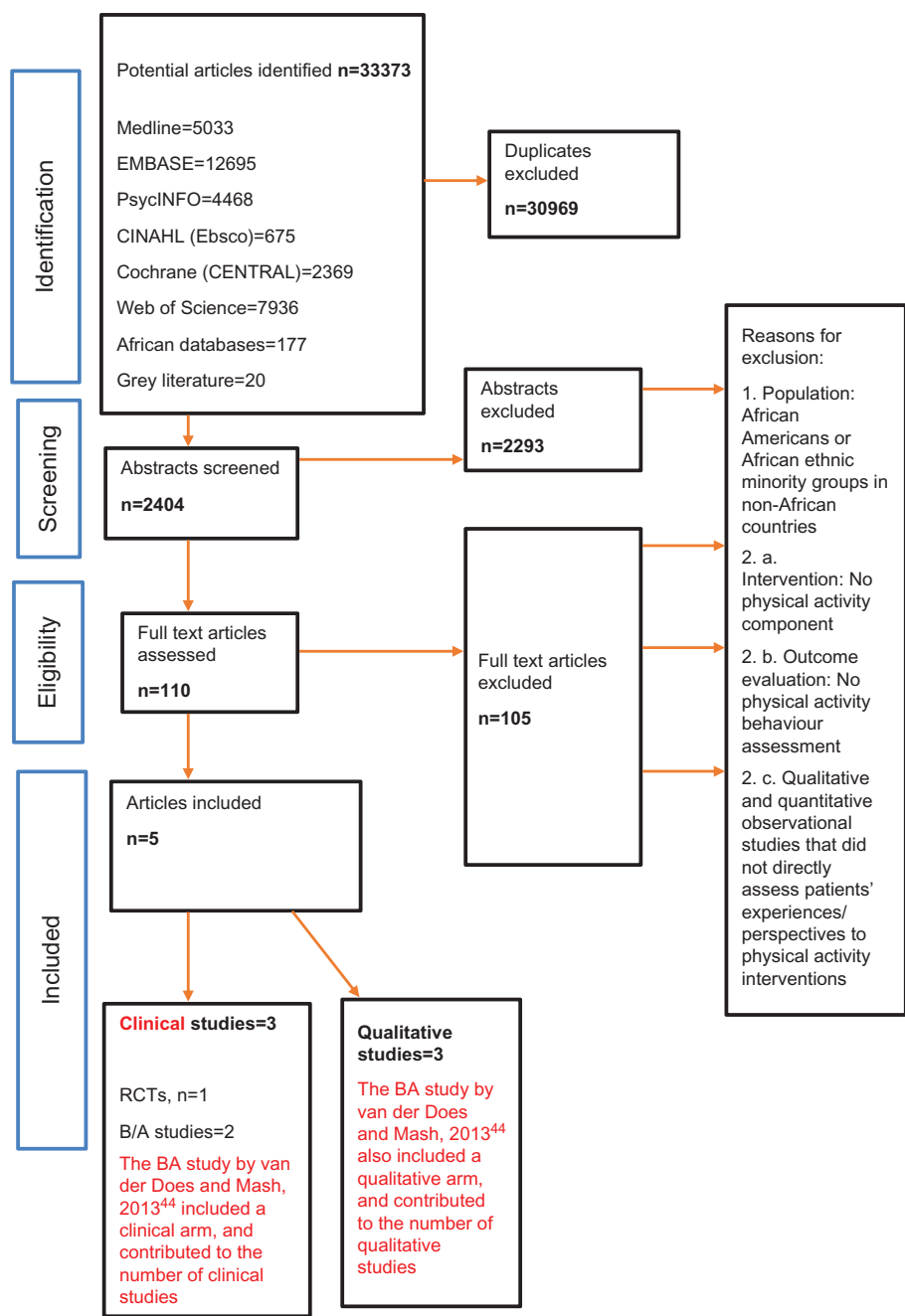


Figure 1. Flow diagram of the selection process of studies for inclusion in the review.

Data analyses

Three-stage analyses were carried out using a mixed-studies synthesis design—the sequential explanatory synthesis.⁴³

First stage: narrative synthesis of clinical studies

Different methods and outcomes employed by studies precluded meta-analysis. There was varied intervention content, objectives, study designs, study quality, sample size, gender,

age, outcome measures and contexts. Clinical studies were therefore summarized using narrative synthesis to determine the effects of interventions on physical activity behaviour. For comparability of results from RCTs and B/A studies, within-group pre- and post-physical activity-change score-effect sizes were calculated for each clinical study.

The overall strength of evidence was judged using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach as described in the review protocol.³⁵

Table 1. Characteristics of clinical studies

Authors	Country	Study design	Study quality MMAT	Chronic condition	Sample size	Gender	Age (years)	Socio-economic status	Outcomes measured	Scales and methods of measurement	Setting
Debussche et al. (2012) ⁴⁰	Reunion Island	RCT	***	T2DM	I=206 C=192	Female=58.8%	Range: ≤18 Mean I=53.8 Mean C=53.7	Mixed	Primary: HbA _{1c} Secondary: BMI, waist circumference, fat mass, SBP, DBP, current treatment, diet, physical activity, fasting blood glucose, lipid profile-plasma total cholesterol, high density lipoprotein (HDL) cholesterol, triglycerides, attendance rate	Continuous (ratio, ordinal) Physical measurements Questionnaire	Mixed
van der Does and Mash (2013) ⁴¹	South Africa	Mixed method (non-controlled B/A and QUAL)	***	T2DM	I=84	Female=81%	Mean: 51.6	Low	Diet (general), diet (specific), physical activity, foot care, medication adherence, ability to teach others	Ordinal (analysed as continuous) and Binary Questionnaire	Rural
Rotheram-Borus et al. (2012) ⁴²	South Africa	Non-controlled B/A	**	T2DM*	I=22	Female=100%	Range: 21–74 Mean: 53	Low	Primary: uptake of text messaging, attendance at small group meetings, perceived quality of the intervention and acceptability of a mobile phone-based peer support intervention. Secondary: Average blood glucose level, BMI, SBP, DBP, emotional distress, coping, weekly exercise and the number of steps taken daily	Continuous (ratio, ordinal) and frequencies Physical measurements, Phone, Questionnaire	Urban

MMAT, mixed-methods appraisal tool (**50%; ***75%); I, intervention/experimental group; C, control group; RCT, randomized controlled trial; B/A, before and after; QUAL, qualitative; T2DM, Type 2 diabetes mellitus; T2DM*, only one subject had Type 1 diabetes mellitus; BMI, body mass index; HbA_{1c}, fasting glycosylated haemoglobin; SBP, systolic blood pressure; DBP, diastolic blood pressure.

Second stage: thematic synthesis of qualitative studies

Qualitative studies were reviewed to explore patients' experiences/perspectives of these interventions (no eligible observational study was identified). NVivo version 10 software (QSR international, Melbourne, Australia) was used.

Thematic synthesis is the method that fits with the aims of this review⁴⁴ because it answers questions relating to intervention need, appropriateness, acceptability and effectiveness, without compromising on key principles of systematic reviews.^{45,46} We used descriptive line-by-line coding to generate categories. Categories were then organized into related areas to construct descriptive themes. Finally, higher order analytical themes were developed through interpretation and abstraction of descriptive themes. Each stage of qualitative synthesis was validated by the review team comparing generated categories/themes with primary results.

Third stage: comparative analysis of clinical and qualitative studies

A comparative analysis was done to determine to what extent clinical studies addressed patients' needs and concerns identified in the qualitative studies.

Results

Overview of the searches

Figure 1 shows a detailed overview of the searches. Table 1 shows the characteristics of clinical studies and quantitative aspects of mixed-methods studies. Table 2 shows characteristics of qualitative studies and qualitative aspect of mixed-methods studies; Table 3 shows a standardized intervention description (intervention components) provided by the reviewers. Table 4 displays a comparative analysis of clinical and qualitative results, while the Supplemental Table shows a description of the intervention and outcomes reported by the primary authors.

Five independent articles, all of which involved patients with diabetes mellitus, were eligible: one RCT=Debussche et al. (2012);⁴⁰ one B/A study=Rotheram-Borus et al. (2012);⁴² one mixed-methods B/A study=van der Does and Mash (2013);⁴¹ two qualitative studies=Serfontein and Mash (2013)⁴⁷ and Balcou-Debussche and Debussche (2009)⁴⁸ (Tables 1 and 2). van der Does and Mash⁴¹ included clinical and qualitative arms, and therefore contributed to the number of both clinical and qualitative studies. The reference list of the relevant systematic review⁴⁹ did not yield additional studies.

Overview and methodological quality of included studies

Identified studies were published between 2009 and 2014 from South Africa (three studies) and Reunion (two studies). Reunion is in the African search filter due to its geographical location in Eastern Africa in the Indian ocean, east of Madagascar and southwest of Mauritius.³⁵ There were a total of 517 participants, ranging between 13 and 398 participants in individual studies. The proportion of women ranged from 38% to 100%. Settings

included rural (n=1), urban (n=2) and mixed (n=2; Tables 1 and 2).

For the methodological quality of the clinical studies, Debussche et al. (2012)⁴⁰ and van der Does and Mash⁴¹ were 75%, whereas quality was 50% in Rotheram-Borus et al. (2012)⁴² (Table 1). Debussche et al. (2012)⁴⁰ had a usual care control group. There were no control groups in van der Does and Mash⁴¹ and Rotheram-Borus et al. (2012)⁴² (supplemental table). Debussche et al. (2012)⁴⁰ was low risk for bias due to blinding of patients, complete outcome data and non-selective reporting, but it was unclear whether outcome assessors were blinded. van der Does and Mash,⁴¹ and Rotheram-Borus et al. (2012)⁴² were high risk for bias due to lack of random sequence generation, allocation concealment and blinding of outcome assessment, but they were low risk for bias due to incomplete outcome data and selective reporting. For the methodological quality of the qualitative studies, Serfontein and Mash (2013)⁴⁷ and Balcou-Debussche and Debussche (2009)⁴⁸ were 100% (Table 2). The quality of van der Does and Mash⁴¹ is denoted above.

Interventions in clinical studies

Debussche et al. (2012)⁴⁰ combined educational sessions and physical activity performance. van der Does and Mash⁴¹ had interventions only comprising interactive educational sessions. Rotheram-Borus et al. (2012)⁴² combined psychoeducational sessions, with mobile phone-based peer support, and was the only study with a theory-based intervention—power-to-prevent programme (supplemental table).

Table 3 shows a total of 18 BCTs. Intervention providers included peers,⁴² educators,⁴⁰ health promoters,⁴¹ dieticians,^{40,41} nurses,⁴⁰ exercise physiologists⁴⁰ and a medical doctor.⁴¹ The duration of interventions ranged between 4 weeks and 9 months. Intervention frequency ranged from once every week to once every 3 months. Session duration was around 1, although it was unspecified in most studies. Interventions were delivered in primary care centres^{41,42} or a tertiary hospital⁴⁰ (Table 3).

Outcome measurement

All studies utilized different self-report measures and none used objective or direct measures of physical activity.

In Debussche et al. (2012),⁴⁰ physical activity was assessed by the adaptation of a validated questionnaire,⁵⁰ calculating occupational and home physical activity scores based on the mean of five items dealing with the frequency of sitting, standing, walking, lifting heavy loads and sweating during activity. Additionally, sports activity was measured by combining intensity (millijoules per hour, i.e. energy expended during the activity per hour), time spent each week (number of hours doing the physical activity per week) and perseverance (number of months that the activity was performed per year). van der Does and Mash⁴¹ used the Summary of validated Diabetes Self-Care Activities questionnaire that assessed diet, physical activity, medication taking, blood-glucose testing, foot care and smoking.⁵¹ The physical activity subscale included two items: 'on how many of the last SEVEN DAYS did you participate in at least 30 minutes of physical activity? (Total minutes of continuous

Table 2. Characteristics of qualitative and mixed-methods studies

Authors	Country	Study designs	Study quality MMAT	Objectives	Chronic condition	Sample size	Gender	Age (years)	Socio-economic status	Setting	Data collection format/ analyses	Themes	Authors' conclusions
Serfontein and Mash (2013) ⁴⁷	South Africa	Qualitative description	****	To explore patients' experience of an educational programme (MI) To explore change in self-care activities, including diet, medication, physical activity and foot care.	Diabetes	13	Female= 92.3%	Range: 41–68	Low	Urban	In-depth interviews/ Framework method	Experiences of the educational programme (1) Information on diabetes (2) Organization of the educational sessions (3) Educational material (4) Communication style (5) Session structure (6) Competency of health promoters Changes in self-care activities Diet Physical activity Medication Foot care Other self-activities	Study supports wider implementation of this programme, following considerations of patient recommendations
van der Does and Mash, (2013) ⁴⁶	South Africa	Mixed methods (non-controlled B/A with qualitative description)	***	To evaluate the Take Five School (TFS) group education programme	T2DM	95	Female= 81.0%	Mean: 51.6	Low	Rural	Focus group interviews, Individual interviews/ Framework method	(1) Strengths of the programme (2) Weaknesses of the programme (3) Opportunities (4) Threats	Study supports the effectiveness of group diabetes education in primary care context
Balcou-Debussche and Debussche, (2009) ⁴⁸	Reunion	Ethnography	****	To assess the effects of self-management education on the individual practices of patients after discharge from specialized diabetes units	T2DM	42	Female= 67.0%	Mean: 53.5	Mixed	Mixed	Ethnographic observations in the hospital and at home, semi-structured interviews/ content analysis of recurrent themes	(1) The Hospital System: A Symbolic (2) Frame of Reference (3) Suspension of reality (4) From hospital to home	

B/A, before and after clinical trial; MMAT, mixed-methods appraisal tool (***75%; ****100%); DM, diabetes mellitus; T2DM, type 2 diabetes mellitus.

Table 3. Intervention components in the clinical studies

Authors	Behaviour change techniques (BCTs)*	Format	Provider/ training	Intervention duration	Session duration/ Frequency	Context within which intervention was delivered
Debussche et al. (2012) ⁴⁰	Shaping knowledge (information provision), restructuring the physical and social environment, behavioural rehearsal/ practice, habit formation, self-monitoring of behaviour, goal setting (behaviour), feedback on behaviour, problem-solving, action planning, health consequences, regulate negative emotions, prompts/cues and action planning (including implementation intentions)	Face-to-face, group-based	Nurse, dietician, Creole-speaking educators, exercise physiologist/not stated	9 months	Not stated/ Once every 3 months	Hospital
van der Does and Mash (2013) ⁴¹	Shaping knowledge (information provision), health consequences, self-monitoring of behaviour and outcome of behaviour, action planning, problem-solving, goal setting (behaviour), identification of self as role model, pharmacological support and action planning (including implementation intentions).	Face-to-face, group-based	Dietician, Health promoter, a doctor/not stated	4 weeks	One hour/ Once a week	Primary health care clinic
Rotheram-Borus et al. (2012) ⁴²	Shaping knowledge (information provision), social support (practical), social support (general), social support (emotional), problem-solving, regulate negative emotions, health consequences, feedback on behaviour, goal setting (behaviour), action planning, instruction on how to perform a behaviour, self-monitoring of outcome of behaviour, prompts/cues	Face-to-face, group-based, telephone-based	Peers/role models (with diabetes)/ training provided	12 weeks	Not stated/ Once a week	Local health clinic

*Labelling was done by the reviewers.

Table 4. Comparative analysis of clinical and qualitative results

Intervention components suggested to be important to patients from qualitative studies	Clinical studies where found
Procedure-related components	
Intervention format: combined group and individual face-to-face sessions in Theme 2 : addressing environmental and social barriers	None identified
Intervention provider: health professionals trained in behaviour change techniques in Theme 3 : stimulating and supporting physical activity behaviour change	None identified
Health facility adapted for chronic disease management in Theme 3 : stimulating and supporting physical activity behaviour change	None identified
Intervention duration: ongoing with possibility of repeat sessions in Theme 3 : stimulating and supporting physical activity behaviour change	None identified
Behaviour change techniques (BCTs)	
Theme 1 : increasing biomedical understanding of chronic condition (BCT: shaping knowledge)	Debussche et al. ⁴⁰ van der Does and Mash ⁴¹ Rotheram-Borus et al. ⁴²
Used self-help educational materials	None identified
Theme 2 : Addressing environmental and social barriers	
Addressed environmental and social barriers (BCTs: restructuring the physical and social environments)	Debussche et al. ⁴⁰
Social support, coping and stress management (BCTs: regulate negative emotions, social support [practical], social support [general] and social support [emotional])	Rotheram-Borus et al. ⁴²
Theme 3 : Stimulating and supporting physical activity behaviour change	
Emphasized complications of chronic condition to increase motivation to improve physical activity (BCT: health consequences)	van der Does and Mash ⁴¹
Actual performance of physical activity within sessions (BCTs: behavioral rehearsal/practice and habit formation)	Debussche et al. ⁴⁰

activity)’ and ‘on how many of the last SEVEN DAYS did you participate in a specific exercise session other than what you do around the house or as part of your work?’. Rotheram-Borus et al. (2012)⁴² measured physical activity using the number of steps taken per day.

Effects of interventions on physical activity

Mixed effects of interventions on physical activity behaviour were found.

In Debussche et al. (2012)⁴⁰ (n=398; quality=75%; validated habitual physical activity questionnaire incorporating home, leisure, occupational and sports measure⁵⁰), the physical activity intervention involving combined interactive educational sessions about diabetes and performance of physical activities during workshops, significantly increased sports activity at 1-year follow-up (effect size=0.4). However, occupational and home physical activity did not differ from baseline. Patients were then split into intervention and control groups. Subsequent quarterly outpatients counselling visits provided to the intervention group did not provide additional benefit over the control group (supplemental table).

In van der Does and Mash⁴¹ (n=84; quality=75%; physical activity subscale of validated Diabetes Self-Care Activities questionnaire which assessed housework, work activities and specific exercise⁵¹), the physical activity intervention only comprising interactive educational sessions about physical activity significantly increased continuous physical activity (at least 30 min) outside of home, and occupational activities immediately

after the 4-week intervention (effect size=0.6) (supplemental table).

The intervention in Rotheram-Borus et al. (2012)⁴² (n=22; quality=50%; physical activity measure=number of steps taken daily), combined psychoeducational sessions about physical activity with mobile phone-based peer support, and did not increase physical activity, but the reduced number of steps from 1931.8 at baseline to 1559.8 at 6 months follow-up (effect size=-0.2) (supplemental table).

Experiences and perspectives of patients to physical activity interventions

Three themes emerged:

Increasing biomedical understanding of the chronic condition—diabetes

Organized, simple and comprehensive structured educational programmes, covering all aspects of the patients’ chronic condition⁴¹ including management strategies⁴⁷ were important to participants.⁴⁷ Recapping previous educational sessions enhanced recall.⁴⁷ Communication styles incorporating motivational interviewing principles such as empathy, liveliness, inclusiveness⁴⁷ and collaborative group sessions enhanced understanding.^{47,48}

Self-help educational materials, such as handouts, flip charts, booklets and posters, enhanced understanding and recall, and made the sessions more interesting.^{41,47,48} Health education

delivered within mainstream healthcare settings bestowed value and legitimacy to self-management recommendations.⁴⁸

Addressing physical and social environmental barriers

Adverse environmental conditions and social stressors may limit physical activity despite sufficient biomedical knowledge of a chronic condition.^{41,47,48} Examples include unsafe environments, poor timing of physical activity,⁴⁷ family problems and stressful work conditions.^{41,47,48}

Social support facilitated through the health professional and other patients in group sessions may mitigate stressors ('... when I sit with other people, I feel better'^{41,47}). Combined group and individual sessions may be superior to group sessions, due to privacy concerns.⁴¹

Strategies stimulating/supporting physical activity behaviour change

Physical activity behaviour may be influenced by motivational interviewing techniques, such as eliciting talk about change, using enthusiastic communication styles and encouraging group solutions.^{41,47} Emphasizing chronic disease complications may further increase the motivation to improve physical activity.⁴¹

Actual performance of the desired physical activity may increase physical activity through increasing self-efficacy.⁴¹ Untrained health professionals operating within a biomedical health care system were more likely to be impatient with patients, damaging the patient–health professional relationship.^{41,48} These health professionals undervalue the skills needed for administering behaviour change interventions, lack ownership, doubt the effectiveness of programmes and are poorly motivated, which might be alleviated by providing ongoing training and support.⁴¹

Potentially limiting behaviour changes were inadequate venues or programmes that were not integrated into routine care.⁴⁷ Unintegrated programmes meant that there were conflicting opinions about who should lead interventions,⁴¹ and timing problems.⁴⁷ Patients wanted ongoing programmes with the possibility of repeat sessions, with specific dates and times that aligned with other hospital appointments/daily activities, and programmes delivered by specifically trained health professionals.⁴¹

The extent to which interventions addressed patients' concerns

Table 4 shows that no clinical study targeted any of the procedure-related intervention components identified in the qualitative studies. All clinical studies^{40–42} adopted techniques that 'increased patients' biomedical understanding of their chronic condition' (BCT: shaping knowledge). However, none used self-help educational materials.

Debussche et al. (2012)⁴⁰ addressed both physical and social environmental barriers (BCTs: restructuring the physical and social environments). Rotheram-Borus et al. (2012)⁴² addressed only social factors (BCTs: social support [practical], social support [general], social support [emotional], regulate negative emotions) (Table 4).

In Debussche et al. (2012),⁴⁰ there was performance of physical activity during educational sessions (BCTs: behavioural rehearsal/practice and habit formation). In van der Does and Mash,⁴¹ complications of the chronic condition were emphasized (BCT: health consequences; Table 4).

Discussion

Summary of main results

All studies involved patients with type 2 diabetes mellitus, with one patient having type 1 diabetes mellitus in Rotheram-Borus et al. (2012).⁴² No study of patients with hypertension, obesity or chronic musculoskeletal condition could be included in this review. Therefore, the results can only be very tentative, based on the limited evidence available for this review.

Moderate quality of evidence suggested that exercises and sports performance not associated with home and occupational activities were increased in the long term—1 year (in terms of frequency per week, energy expended per hour, number of hours per week and number of months the activities were performed per year). Low quality of evidence suggested that exercises and sports activities increased in the short term—immediately after a 4-week intervention (in terms of total minutes of continuous activity).^{40,41} There was conflicting evidence on the effects of interventions on home and occupational physical activities.^{41,42}

This review could not confirm the BCTs responsible for improvements in physical activity because primary clinical studies did not investigate this. Despite possible links between engagement with physical activity interventions and physical activity behaviour change,⁵² no study examined this. Furthermore, it was not possible to undertake a meta-regression to determine the intervention components for improving physical activity due to the limited number and low quality of studies.

A meta-regression of clinical studies in high income countries found that interventions that combined self-monitoring of behaviour with at least one of four other self-regulatory techniques, i.e. prompting intention formation, providing feedback on performance, prompting specific goal setting and review of behavioural goals, were significantly more effective than other interventions in improving physical activity and healthy eating.⁵³ However, that meta-regression did not thoroughly investigate the impact of procedure-related intervention components.⁵³ The impact of theoretical constructs on physical activity behaviour could not be ascertained in the current review because only Rotheram-Borus et al. (2012)⁴² had a theory-informed intervention and did not measure theoretical constructs.⁴² Evidence suggests that no theory is superior in influencing physical activity, although interventions based on the social cognitive theory produced the greatest effect sizes.⁵⁴ However, none of the studies was from an African setting.

The BCTs in Debussche et al. (2012)⁴⁰ and van der Does and Mash,⁴¹ which improved physical activity in the long and short term, respectively, included self-monitoring of behaviour and at least one other self-regulatory technique. Providing feedback on behaviour and prompting goal setting were two other self-regulatory techniques in Debussche et al. (2012).⁴⁰ Prompting goal setting was one other self-regulatory technique in van der

Does and Mash.⁴¹ The techniques in Rotheram-Borus et al. (2012),⁴² did not improve physical activity and did not include self-monitoring of behaviour, but provided feedback on behaviour and prompted goal setting.

In the comparative analysis in Table 4, Debussche et al. (2012)⁴⁰ was the only clinical study that reported physical activity changes in the long term (1 year), and used techniques addressing all three themes identified in the qualitative studies. van der Does and Mash,⁴¹ which reported increases in physical activity in the short term, addressed two of the three themes, but did not tackle environmental and social barriers. Rotheram-Borus et al. (2012),⁴² which found no improvements in the number of steps taken daily, utilized BCTs that targeted two of the three themes, but did not use BCTs 'stimulating/supporting physical activity' (Table 4).

No intervention addressed any procedure-related intervention component identified as important in the qualitative studies (Table 4). Fidelity was not assessed in any study, and untrained health professionals may have limited the extent to which identified BCTs were delivered with fidelity. Although Debussche et al. (2012)⁴⁰ mentioned training health professionals in counselling, there were no details recorded of this training, and no assessment of intervention fidelity.⁴⁰

Although some clinical studies on obesity, type 2 diabetes mellitus and hypertension identified during this review included behaviour change intervention components, and measured clinical outcomes, such as blood pressure and fasting blood glucose, they were excluded because physical activity was not measured in any of them (Figure 1). This highlights the key emphasis on biomedical, rather than behavioural outcomes in Africa which may prevent successful behaviour change. No study on chronic musculoskeletal pain was identified at all, highlighting the limited focus of physical activity behaviour change for chronic musculoskeletal pain in Africa. Systematic reviews of studies conducted in high income countries suggest that exercise supervision, participation in an exercise programme, accuracy of exercise performance, participation in a motivational behaviour change programme, refresher/booster/follow-up sessions, supplementing face-to-face verbal instructions with other audio/visual/written material, reward and punishment strategies, goal setting, feedback, group discussion, problem-solving to overcome barriers to adherence, self-monitoring through the use of an exercise plan/exercise contract/exercise logbook, education about pathology and how to manage/cope with symptoms, assessing community resources/social networks, effective communication, reinforcement and feedback, and cognitive behavioural strategies may improve physical activity in patients with chronic musculoskeletal pain.^{55–58} Most of these components were identified in this review, suggesting their significance in other chronic conditions.

Measurement of physical activity

Lack of direct or objective physical activity measurement in studies is a potential weakness.³⁴ Although self-reported physical activity measures are the most practical and cost-effective approach to physical activity measurement, including both direct/objective and self-report measures, they may account for

the weaknesses of both measurement types. For instance, high sensitivity to small errors and measurement of non-physical activity energy expenditure are associated with direct/objective physical activity measures.³⁴ Greater estimation of high intensity/vigorous physical activity than low to moderate physical activity, plus recall and social desirability bias are associated with self-reported physical activity measures.³⁴

The physical activity measurement in Debussche et al. (2012)⁴⁰ acknowledged work activities, housework, sports and leisure activities within the context of frequency, intensity, duration and overall physical activity over time, concurring with the WHO recommendations.²⁴ In van der Does and Mash,⁴¹ physical activity was measured within the context of housework, work activities and specific exercise, taking into consideration frequency and duration, but not intensity. In Rotheram-Borus et al. (2012),⁴² the measurement of physical activity only using the number of steps taken per day, was the least validated of the measures as it did not consider duration, intensity and physical activity that does not involve taking steps.

Strengths and limitations

This review combined the strengths inherent in quantitative and qualitative studies, thereby compensating for their respective limitations. However, the generalizability of findings is limited by both the quality of evidence and paucity of studies conducted in only a few countries. Only studies of patients with diabetes mellitus in South Africa and Reunion were eligible, limiting generalizability. Although Reunion is classified within Eastern Africa, it is a French overseas territory, also including Europeans, Indians and Chinese, which is a limitation. Many interventions were not physical activity-specific, and were very complex, containing dietary and pharmacological elements, which may have influenced outcomes. Important intervention components were derived from a few studies. Suboptimal physical activity measurement could have influenced outcomes.

Conclusions

Limited evidence suggested that physical activity, unrelated to home and occupational activities, was improved and that self-monitoring of physical activity may be associated with this improvement. Inconclusive evidence was found for home and occupational physical activity. Patients appreciated BCTs that improved chronic disease knowledge, addressed environmental barriers and supported physical activity. Procedure-related components, including health professional training and adequate health facilities, were not tackled by any study despite being important to patients.

High quality RCTs in Africa should test interventions delivered by trained health professionals, and focusing only on physical activity, assessed with validated measures, and link this to short- and long-term behavioural/biomedical outcomes. This will improve the evidence for these interventions in Africa. These behaviour-change interventions need to be patient-centred by being responsive to patients' concerns, preferences and needs.

Supplementary data

Supplementary data are available at International Health Online (<http://inthealth.oxfordjournals.org>).

Authors' contributions: CNI-C conceived the study, developed the search strategy, pilot searched the databases, selected studies and synthesized data. CNI-C, ELG, APK and IOS contributed to further development of the review protocol, selection criteria, quality assessment, the risk of bias assessment, data extraction and analyses. CNI-C prepared the draft manuscript. All authors contributed to a critical revision of the manuscript. CNI-C prepared the final manuscript. All authors approved the final manuscript. CNI-C is the guarantor.

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